contrast, the most straightforward reading of string theory says that the extra space dimensions are every bit as real as the three we know about.

20. String theory experts (and those who have read *The Elegant Universe*, Chapter 12) will recognize that the more precise statement is that certain formulations of string theory (discussed in Chapter 13 of this book) admit limits involving eleven spacetime dimensions. There is still debate as to whether string theory is best thought of as fundamentally being an eleven spacetime dimensional theory, or whether the eleven dimensional formulation should be viewed as a particular limit (e.g., when the string coupling constant is taken large in the Type IIA formulation), on a par with other limits. As this distinction does not have much impact on our general-level discussion, I have chosen the former viewpoint, largely for the linguistic ease of having a fixed and uniform total number of dimensions.

Chapter 13

1. For the mathematically inclined reader: I am here referring to *conformal* symmetry—symmetry under arbitrary angle-preserving transformations on the volume in spacetime swept out by the proposed fundamental constituent. Strings sweep out two-spacetime-dimensional surfaces, and the equations of string theory are invariant under the two-dimensional conformal group, which is an *infinite* dimensional symmetry group. By contrast, in other numbers of space dimensions, associated with objects that are not themselves one-dimensional, the conformal group is finite-dimensional.

2. Many physicists contributed significantly to these developments, both by laying the groundwork and through follow-up discoveries: Michael Duff, Paul Howe, Takeo Inami, Kelley Stelle, Eric Bergshoeff, Ergin Szegin, Paul Townsend, Chris Hull, Chris Pope, John Schwarz, Ashoke Sen, Andrew Strominger, Curtis Callan, Joe Polchinski, Petr Hořava, J. Dai, Robert Leigh, Hermann Nicolai, and Bernard deWit, among many others.

3. In fact, as explained in Chapter 12 of *The Elegant Universe*, there is an even tighter connection between the overlooked tenth spatial dimension and *p*-branes. As you increase the size of the tenth spatial